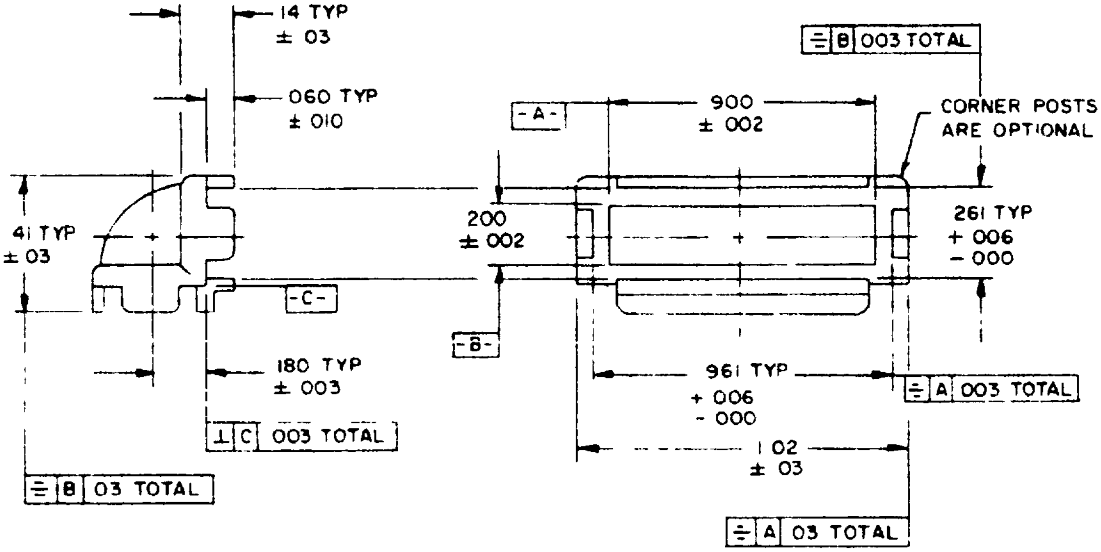
MILITARY SPECIFICATION SHEET

WAVEGUIDE ASSEMBLY, RIGID, CAST, 90-DEGREE E-PLANE BEND, REDUCED HEIGHT, RADIUS-BACK MITER, FULL CORRAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the waveguide assembly described herein shall consist of this document and the latest issue of Specification MIL-W-3970



INCHES	MM	INCHES	MM	INCHES	MM
002	05	060	1 52	41	10 4
003	08	14	3.6	900	22 86
006	15	180	4 57	961	24 41
010	25	200	5 08	1 02	25 9
03	8	261	6 63		

NOTES
1. Dimensions are in inches
2 Metric equivalents are given for general information only and are based upon 1 00 inch = 25 4 mm

FIGURE 1 Dimensions for reduced height, radius-back miter, and full corral configuration

[illegible]

Physical characteristics:
Dimensions and configuration: See figure [illegible].
Operating temperature range: -55° to +3[illegible]°C.
Body material: [illegible]7.2.1, temper [illegible], aluminum alloy per QQ-A-6[illegible], or [illegible]712.1 per [illegible].
Surfaces (sandblasting after machining permissible):
Internal: [illegible] microinches per ANSI B46.1.

External: 125 microinches per ANSI B46.1

Fungus resistance: Materials nutrient to fungus shall not be used.

Pressurization: 45 lb_f/in^2 gage maximum.

Penetrant inspection: No indication of any cracks at any location when tested per MIL-I-6866, type [illegible], method [illegible].

Electrical characteristics:
Frequency range - [illegible]2 to 12.4 GHz.
VSWR: 1.05:1 maximum.
Power:
Peak - 250 kW.
Average - 10 kW.

Part number: M39[illegible]/25-0[illegible]

NOTE: The mating waveguides are 1/2 height of X-band waveguides, with a nominal wall thickness of .030 inch.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Army - MI,
Navy - EC, OS
Air Force - 11, 17, 99
DLA - ES

User activities:
Army - AR
Navy - MC, CG, AS
Air Force - 14, 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5985-0881-5)

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL	OMB Approval No 22 R255

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DOCUMENT IDENTIFIER AND TITLE MIL-W-3970/25

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

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A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

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	DATE

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